# Import Data

In [1]:
```
import pandas as pd
```

In [28]:
```
data = pd.read_csv(r'D:\Downloads\Udemy-PY\Covid_19_data.csv')
```

In [3]:
```
data.head()
```

Out[3]:

| | Date | State | Region | Confirmed | Deaths | Recovered |
|---|---|---|---|---|---|---|
| **0** | 4/29/2020 | NaN | Afghanistan | 1939 | 60 | 252 |
| **1** | 4/29/2020 | NaN | Albania | 766 | 30 | 455 |
| **2** | 4/29/2020 | NaN | Algeria | 3848 | 444 | 1702 |
| **3** | 4/29/2020 | NaN | Andorra | 743 | 42 | 423 |
| **4** | 4/29/2020 | NaN | Angola | 27 | 2 | 7 |

## Checking Metadata

In [4]:
```
data.shape
```

Out[4]:
```
(321, 6)
```

In [5]:
```
data.isnull().sum()
```

Out[5]:
```
Date           0
State        181
Region         0
Confirmed      0
Deaths         0
Recovered      0
dtype: int64
```

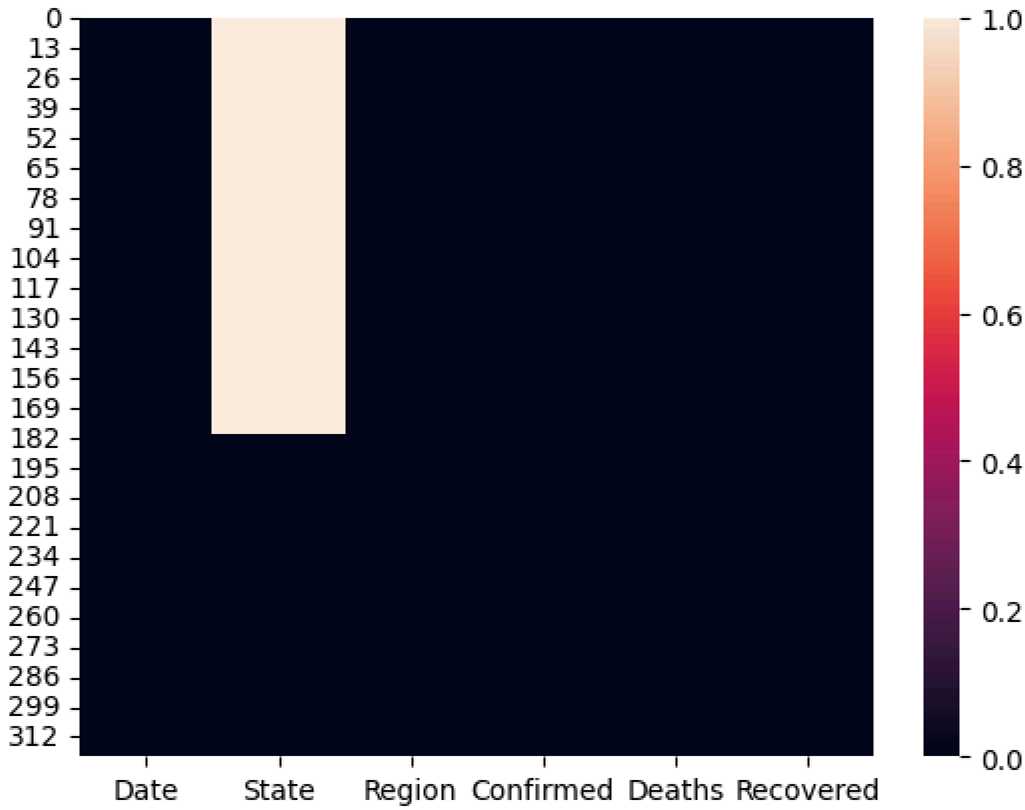
## Showing Null values

In [6]:
```
import seaborn as sns
```

In [7]:
```
import matplotlib.pyplot as plt
```

In [8]:
```
sns.heatmap(data.isnull())
plt.show
```

Out[8]:
```
<function matplotlib.pyplot.show(close=None, block=None)>
```




## 1. Show the number of Confirmed, Death and Recovered cases in each Region

In [9]:
```
data.groupby('Region').sum()
```

Out[9]:

| | Date | State | Confirmed | Deaths | Recovered |
|---|---|---|---|---|---|
| **Region** | | | | | |
| **Afghanistan** | 4/29/2020 | 0 | 1939 | 60 | 252 |
| **Albania** | 4/29/2020 | 0 | 766 | 30 | 455 |
| **Algeria** | 4/29/2020 | 0 | 3848 | 444 | 1702 |
| **Andorra** | 4/29/2020 | 0 | 743 | 42 | 423 |
| **Angola** | 4/29/2020 | 0 | 27 | 2 | 7 |
| **...** | ... | ... | ... | ... | ... |
| **West Bank and Gaza** | 4/29/2020 | 0 | 344 | 2 | 71 |
| **Western Sahara** | 4/29/2020 | 0 | 6 | 0 | 5 |
| **Yemen** | 4/29/2020 | 0 | 6 | 0 | 1 |
| **Zambia** | 4/29/2020 | 0 | 97 | 3 | 54 |
| **Zimbabwe** | 4/29/2020 | 0 | 32 | 4 | 5 |

187 rows × 5 columns

In [10]:

```
#For Deaths seperately:
data.groupby('Region')['Deaths'].sum().head(5).sort_values(ascending = False) #desc
```

Out[10]:

```
Region
Algeria        444
Afghanistan     60
Andorra         42
Albania         30
Angola           2
Name: Deaths, dtype: int64
```

In [17]:

```
data.groupby('Region')[['Confirmed', 'Recovered']].sum()
```

Out[17]:

| Region | Confirmed | Recovered |
|---|---|---|
| Afghanistan | 1939 | 252 |
| Albania | 766 | 455 |
| Algeria | 3848 | 1702 |
| Andorra | 743 | 423 |
| Angola | 27 | 7 |
| ... | ... | ... |
| West Bank and Gaza | 344 | 71 |
| Western Sahara | 6 | 5 |
| Yemen | 6 | 1 |
| Zambia | 97 | 54 |
| Zimbabwe | 32 | 5 |

187 rows × 2 columns

## 2. Remove all records where Confirmed cases are less than 10

In [21]:
```
data[data.Confirmed < 10]
```

Out[21]:

| | Date | State | Region | Confirmed | Deaths | Recovered |
|---|---|---|---|---|---|---|
| **18** | 4/29/2020 | NaN | Bhutan | 7 | 0 | 5 |
| **98** | 4/29/2020 | NaN | MS Zaandam | 9 | 2 | 0 |
| **105** | 4/29/2020 | NaN | Mauritania | 8 | 1 | 6 |
| **126** | 4/29/2020 | NaN | Papua New Guinea | 8 | 0 | 0 |
| **140** | 4/29/2020 | NaN | Sao Tome and Principe | 8 | 0 | 4 |
| **177** | 4/29/2020 | NaN | Western Sahara | 6 | 0 | 5 |
| **178** | 4/29/2020 | NaN | Yemen | 6 | 0 | 1 |
| **184** | 4/29/2020 | Anguilla | UK | 3 | 0 | 3 |
| **192** | 4/29/2020 | Bonaire, Sint Eustatius and Saba | Netherlands | 5 | 0 | 0 |
| **194** | 4/29/2020 | British Virgin Islands | UK | 6 | 1 | 3 |
| **203** | 4/29/2020 | Diamond Princess cruise ship | Canada | 0 | 1 | 0 |
| **272** | 4/29/2020 | Northwest Territories | Canada | 5 | 0 | 0 |
| **284** | 4/29/2020 | Recovered | Canada | 0 | 0 | 20327 |
| **285** | 4/29/2020 | Recovered | US | 0 | 0 | 120720 |
| **288** | 4/29/2020 | Saint Barthelemy | France | 6 | 0 | 6 |
| **289** | 4/29/2020 | Saint Pierre and Miquelon | France | 1 | 0 | 0 |
| **305** | 4/29/2020 | Tibet | Mainland China | 1 | 0 | 1 |

In [26]:
```
data = data[~(data.Confirmed < 10)]
```

In [27]:
```
data
```

Out[27]:

| | Date | State | Region | Confirmed | Deaths | Recovered |
|---|---|---|---|---|---|---|
| **0** | 4/29/2020 | NaN | Afghanistan | 1939 | 60 | 252 |
| **1** | 4/29/2020 | NaN | Albania | 766 | 30 | 455 |
| **2** | 4/29/2020 | NaN | Algeria | 3848 | 444 | 1702 |
| **3** | 4/29/2020 | NaN | Andorra | 743 | 42 | 423 |
| **4** | 4/29/2020 | NaN | Angola | 27 | 2 | 7 |
| **...** | ... | ... | ... | ... | ... | ... |
| **316** | 4/29/2020 | Wyoming | US | 545 | 7 | 0 |
| **317** | 4/29/2020 | Xinjiang | Mainland China | 76 | 3 | 73 |
| **318** | 4/29/2020 | Yukon | Canada | 11 | 0 | 0 |
| **319** | 4/29/2020 | Yunnan | Mainland China | 185 | 2 | 181 |
| **320** | 4/29/2020 | Zhejiang | Mainland China | 1268 | 1 | 1263 |

304 rows × 6 columns

## 3. Find the Region with the Maximum number of Confirmed cases

In [31]:

```
data.groupby('Region').Confirmed.sum().sort_values(ascending = False).head(20)
```

Out[31]:

```
Region
US                  1039909
Spain                236899
Italy                203591
France               166543
UK                   166441
Germany              161539
Turkey               117589
Russia                99399
Iran                  93657
Mainland China        82862
Brazil                79685
Canada                52865
Belgium               47859
Netherlands           38998
Peru                  33931
India                 33062
Switzerland           29407
Ecuador               24675
Portugal              24505
Saudi Arabia          21402
Name: Confirmed, dtype: int64
```

## 4. Find the Region with Minimum number of Deaths

In [37]:
```
data.groupby('Region').Deaths.sum().sort_values(ascending = True).head(20)
```

Out[37]:
```
Region
Cambodia                    0
Bhutan                      0
Dominica                    0
Central African Republic    0
Eritrea                     0
Fiji                        0
Holy See                    0
Mozambique                  0
Macau                       0
Madagascar                  0
Namibia                     0
Mongolia                    0
Nepal                       0
Laos                        0
Grenada                     0
Papua New Guinea            0
Sao Tome and Principe       0
Seychelles                  0
South Sudan                 0
Uganda                      0
Name: Deaths, dtype: int64
```

## 5. List the Confirmed, Deaths and Recovered cases in Pakistan till 29 April 2020

In [42]:
```
data.groupby('Region').get_group('Pakistan').sum() #1
```

Out[42]:
```
Date         4/29/2020
State                0
Region        Pakistan
Confirmed        15525
Deaths             343
Recovered         3425
dtype: object
```

In [45]:
```
data[data.Region=='Pakistan'] #2
```

Out[45]:

| | Date | State | Region | Confirmed | Deaths | Recovered |
|---|---|---|---|---|---|---|
| **124** | 4/29/2020 | NaN | Pakistan | 15525 | 343 | 3425 |

## 6. Sort all data WRT Number of Confirmed cases in Ascending order

In [47]:
```
data.sort_values(by = ['Confirmed'], ascending = True)
```

Out[47]:

| | Date | State | Region | Confirmed | Deaths | Recovered |
|---|---|---|---|---|---|---|
| **284** | 4/29/2020 | Recovered | Canada | 0 | 0 | 20327 |
| **285** | 4/29/2020 | Recovered | US | 0 | 0 | 120720 |
| **203** | 4/29/2020 | Diamond Princess cruise ship | Canada | 0 | 1 | 0 |
| **289** | 4/29/2020 | Saint Pierre and Miquelon | France | 1 | 0 | 0 |
| **305** | 4/29/2020 | Tibet | Mainland China | 1 | 0 | 1 |
| **...** | ... | ... | ... | ... | ... | ... |
| **57** | 4/29/2020 | NaN | France | 165093 | 24087 | 48228 |
| **168** | 4/29/2020 | NaN | UK | 165221 | 26097 | 0 |
| **80** | 4/29/2020 | NaN | Italy | 203591 | 27682 | 71252 |
| **153** | 4/29/2020 | NaN | Spain | 236899 | 24275 | 132929 |
| **265** | 4/29/2020 | New York | US | 299691 | 23477 | 0 |

321 rows × 6 columns

## 7. Sort all data WRT Number of Recovered cases in Descending order

In [48]:

```
data.sort_values(by = ['Recovered'], ascending = False)
```

Out[48]:

| | Date | State | Region | Confirmed | Deaths | Recovered |
|---|---|---|---|---|---|---|
| **153** | 4/29/2020 | NaN | Spain | 236899 | 24275 | 132929 |
| **285** | 4/29/2020 | Recovered | US | 0 | 0 | 120720 |
| **61** | 4/29/2020 | NaN | Germany | 161539 | 6467 | 120400 |
| **76** | 4/29/2020 | NaN | Iran | 93657 | 5957 | 73791 |
| **80** | 4/29/2020 | NaN | Italy | 203591 | 27682 | 71252 |
| **...** | ... | ... | ... | ... | ... | ... |
| **290** | 4/29/2020 | Saskatchewan | Canada | 383 | 6 | 0 |
| **299** | 4/29/2020 | South Dakota | US | 2373 | 13 | 0 |
| **298** | 4/29/2020 | South Carolina | US | 5882 | 231 | 0 |
| **302** | 4/29/2020 | Tennessee | US | 10366 | 195 | 0 |
| **303** | 4/29/2020 | Texas | US | 27257 | 754 | 0 |

321 rows × 6 columns

In [ ]: